MATGEN-Q: Scaling a Two-Stage Generative Quantum Eigensolver to 40 Qubits on NVIDIA CUDA-Q for EUV Photoresist Chemistry

Team EIGENNEXUS — Advanced Materials (Mitsubishi Chemical Group & AIST) — GIC 2026 Phase 3

1. Focus Area and Rationale

Ground-state energy estimation underlies reaction thermodynamics, redox behavior, and excited-state properties throughout materials chemistry, yet classical methods scale exponentially with electron correlation and the gold-standard CCSD(T) costs $O(N^7)$. The Generative Quantum Eigensolver (GQE; Nakaji et al. 2024) takes a different route from VQE: a classical generative transformer proposes the quantum circuits rather than optimizing a variational one, which sidesteps the barren-plateau problem (Tilly et al. 2022; a GQE property established by Nakaji et al. 2024 and adopted here, not re-demonstrated). Until recently, statevector GQE stayed small-scale and QSCI-paired GQE had reached only 32 qubits (Kemmoku, Gao et al. 2026). **Carrying this generative approach into the ~40-qubit, industrially relevant regime, with executed and reproducible results, is the problem Phase 3 addresses.** Our primary executed contribution is quantum-accurate energies for the chemistry where the classical gold standard breaks down. As bonds stretch, single-reference CCSD(T) collapses non-variationally below the exact energy, a confidently wrong result that gives no internal warning; we therefore target the open-shell, multireference transition-metal oxides (CrO, NiO) and EUV-relevant tin oxides (SnO, SnO_2) at 20 qubits, validated against exact diagonalization and third-party HamLib Hamiltonians. Around that core we build the machinery needed to scale: tensor-network plus selected-CI evaluation, MP2 pool compression, and a generator that transfers across system size and chemistry, all aimed at the 40q+ regime, with every claim reproducible and every limitation stated.

Our application target is EUV semiconductor photoresist chemistry (tin-oxo clusters; Sn, Hf, Zr oxides), whose open-shell, multireference character defeats single-reference classical methods. It is also the focus of an active quantum-simulation program by one of this challenge's providers (Kharazi et al., Xanadu & Mitsubishi Chemical, 2026). Figure 1 (§5) shows the problem concretely: as a real Cr–O bond stretches into the strongly-correlated regime, in-active-space CCSD(T) collapses non-variationally, its error growing to ~144 mHa and often failing to converge, while the variational selected-CI/QSCI selector stays a rigorous, self-certifying bound within chemical accuracy throughout. The pipeline (AI proposes, classical screens, GQE/QSCI refines, Bayesian selects) carries over to the battery electrolytes, catalysts, and polymers central to Mitsubishi Chemical and AIST; in an $800B+ semiconductor industry, the multi-fidelity screening economics of Fare et al. (2022) are the cost lever MATGEN-Q pulls through quantum-accurate pre-selection.

Stakeholder relevance. For Mitsubishi Chemical, the payoff is a quantum-accurate, self-certifying filter that runs ahead of costly synthesis and lithographic testing, where single-reference classical methods misjudge strongly-correlated tin-oxo energetics. For AIST/G-QuAT, it is a scalable, reproducible GQE workflow on CUDA-Q that extends the jointly-developed program past its current size limits.

2. Target System and Data Modeling Strategy

Scaling vehicle. We use the H_n chain series (n = 2…20; 4–40 qubits, STO-6G, Jordan–Wigner), the canonical strong-correlation benchmark, because a single bond-length parameter tunes it from weak to strong correlation and so stresses the simulation layer that any scaling claim rests on.

Target chemistry. The same QSCI engine reaches chemical accuracy on real materials chemistry. On Sn-oxide active spaces (Sn ECP-CASCI, validated on H_4 to 0.0000 mHa), it holds chemical accuracy for SnO (16q) and SnO_2 (20q): reproduce.py asserts ≤1.6 mHa, and we observe 0.11–0.46 / 0.13–0.23 across versions and re-runs. To reach a genuine EUV motif rather than a diatomic, we also treat a bridged **Sn–O–Sn tin-oxo fragment (Sn_2O_2 rhombus) to 0.41 mHa at 16q**, and the genuinely open-shell, multireference transition-metal oxides **CrO $^5\Pi$ (0.038 mHa) and NiO $^3\Sigma^-$ (0.197 mHa) at 20 qubits**. These are executed 20-qubit results. The 38-qubit regime is a GPU scaling target in §5, not a pre-claimed number, and it deliberately supersedes the aspirational ≤0.08 mHa @38q figure in our Phase 2 draft, which we no longer claim.

Data integrity. We adopt HamLib (Sawaya et al., Quantum 2024) and built a generation pipeline that replicates its methodology (PySCF, OpenFermion, Jordan–Wigner, STO-6G). **Checked against the published files, our FCI reproduces to 0.00000 mHa (H_2–H_6), and at 28/32/40 qubits our Hamiltonians match HamLib exactly in term count (27,735 / 47,489 / 116,577) and to ~15 significant figures in coefficient magnitude**, differing only by a spectrum-invariant orbital-phase convention.

Accuracy anchors / classical references. We compare against FCI exactly (≤20q) and CCSD(T) near equilibrium. Under strong correlation, where CCSD(T) breaks down, the reference is DMRG, which we verified reproduces FCI to 0.000 mHa at 20q.

3. GQE-Based Approach and Algorithmic Innovation

MATGEN-Q is a two-stage GQE. **Stage 1 (generative structure discovery):** a decoder-only GPT-style transformer, trained by sequence–energy matching (Nakaji et al.), writes circuits as token sequences over a UCC single/double excitation pool. **Stage 2 (continuous refinement):** adjoint-gradient angle optimization converges them to chemical accuracy. Four design choices make it scale:

(1) Tensor-network (MPS) simulation, the GPU scaling enabler; the bond-dimension argument is now demonstrated on CPU. CUDA-Q's tensornet-mps memory scales with entanglement, not 2^n, and we verify that enabling claim with block2 DMRG: **the bond dimension χ needed for chemical accuracy grows only modestly with size (χ≈50/100/400 at 20/28/40 qubits), entanglement area-law-bounded near equilibrium (S_{max} 0.39) and rising into strong correlation (S_{max} 4.43)**. At 40q, then, χ≈400 keeps the MPS footprint (~$n \cdot \chi^2 \cdot d$, megabytes) far below the ~16 TB exact statevector. The 20q/28q GPU runs are executed to chemical accuracy (§5); the full 40q tensornet-mps run remains the primary owed deliverable, de-risked by the measured χ target and the ≈5× faster growth engine.

(2) QSCI energy evaluation, accuracy- and noise-aware. In Quantum-Selected Configuration Interaction (Kanno et al. 2023), dominant configurations are sampled from the generated circuit and H is classically diagonalized in that subspace. Because the device defines only the subspace, the method is intrinsically noise-robust: at 20q with 2×10^6 shots the energy degrades gracefully to ≤3.3 mHa even with 30% of measurements corrupted, and run through CUDA-Q's density-matrix backend with a physical depolarizing channel, QSCI on H_4 holds chemical accuracy up to 5% per-gate noise. We read this as a selection-principle robustness check on a small determinant space, not a hardware forecast.

(3) Operator-pool compression, for efficiency. Ranking the $O(N^4)$ double-excitation pool by active-space MP2 amplitude and keeping the top fraction shrinks the transformer vocabulary with negligible accuracy loss, exactly where random pruning collapses. In a deterministic CI-subspace test (CO/N_2/SiO, 12q), **N_2 retains full-pool accuracy (2.26 mHa) keeping only 25% of doubles (vocabulary 1170→430), against 50.4 mHa for random pruning at the same size (~22×)**; CO holds 3.7 mHa at 40% kept versus 29.7 mHa random.

(4) Distributed hybrid workflow, detailed in §4.

Transfer learning: a generative prior that carries across system size and chemistry. With a canonical, frontier-relative tokenization (each excitation encoded as occupied-depth-below-HOMO and virtual-height-above-LUMO, independent of qubit count), a small system's token vocabulary is a provable subset of a larger one (H_6/12q ⊂ H_8/16q ⊂ H_{10}/20q). One GPT-QE generator trained only on H_4+H_6 (8q+12q) is then deployed zero-shot across 16→56 qubits. **The robust signals are statistical: a ~3.6× tighter selection spread than random (≈1.4 vs ≈5.1 mHa across-seed SD) and 43/48 size×seed paired wins at 8 seeds (binomial $p<0.0001$)**, reliable selection rather than a lottery. The per-size mean advantage is positive at every size and stays significant (7/8 wins) through 56q, narrowing only in magnitude (+7.1→+3.9 mHa, 16→56q). Interpretability suggests the policy has learned real chemistry: trained on energy alone, it recovers the MP2 double-excitation amplitude hierarchy (Spearman $\rho=0.31$, $p<0.002$; 4 of its top-8 doubles are MP2's). This is the *train-small*, *deploy-large* idea. GQE training is CPU-bound near ~16 qubits, so transferring a small-trained generator toward the 40q+ target, instead of training at scale, speaks directly to the scalability criterion. What makes 56q reachable on CPU is a determinant-space evaluation (each excitation maps by bitmask to a determinant, diagonalized via Slater-Condon, validated to 0.0000 mHa against Jordan–Wigner) that needs no 2^n statevector. This is a *selected-CI proxy* for circuit-sampled QSCI: relative at a small fixed budget (K=96, where the generator beats random at every budget up to K=96 but converges with it by K=128), not absolute chemical accuracy.

Cross-chemistry transfer (honest, partial). Deployed to oxide active spaces, the same generator beats random on 3 of 6 targets (CO-12q/SiO/CO-20q), ties BeO, and loses on SnO. Target-specific MP2 is strongest on every target. So cross-size transfer is robust while cross-chemistry works only when the target resembles training: a zero-cost prior, not a universal one. A same-size conditional-GQE variant (cf. Minami, Nakaji et al. 2025) gave only a within-noise gain, which we also report as a negative.

4. Hybrid Architecture

The classical transformer trains on GPU (PyTorch), while circuit evaluations are dispatched across GPUs through CUDA-Q's mqpu in an asynchronous generate→evaluate→update loop. Quantum resources handle state preparation and energy estimation (MPS/QSCI); classical resources handle generation, optimization, and active-space selection. **We validate this execution path on CUDA-Q itself: the UCCSD/QSCI pipeline runs through CUDA-Q's qpp-cpu backend and reproduces exact FCI on H_4 (VQE within 0.02 mHa via cudaq.observe, QSCI within chemical accuracy via cudaq.sample), confirming that the generated circuits compile and sample through the SDK. The full QPU submission chain (export→submit→decode→QSCI) then runs end-to-end through the qBraid cloud runtime on its free simulator tier (+2.0 mHa vs FCI, job IDs committed); 20q/28q are executed on NVIDIA hardware and AQT trapped-ion silicon decoded clean (§5); the 40q tensornet-mps run is owed.**

5. Phase 3 Execution and Results

Verified results (CPU). A single command (*reproduce.py*) runs 26 automated checks: 17 re-executions plus 9 committed-artifact audits, among them the pre-registration SHA-256, the AQT trapped-ion decode, and a cost audit that re-derives the program spend from published pricing and committed shot/uptime configs. The result is **26/26 PASS**, and every value traces to a committed result file (Table 1).

System	Qubits	Quantum (GQE/QSCI)	Classical ref.	Notes
H_2/H_4/H_6	**4/8/12**	**0.000 / 0.009 / 0.297 mHa**	**FCI (exact)**	**two-stage GQE (UCCSD)**
H_6 GQE→QSCI	**12**	**1.05 mHa (51→1.05, ~50×)**	**FCI**	**measured pipeline**
H_{10}/H_{14}	**20/28**	**0.57 / 1.21 mHa**	**FCI / CCSD(T)**	**2,401 / 18,201 dets**
CrO $^5\Pi$ / NiO $^3\Sigma^-$	**20**	**0.038 / 0.197 mHa**	**CASCI (exact)**	**open-shell multiref.**
SnO / SnO_2	**16/20**	**chem. acc. ≤1.6 (obs. 0.11–0.46 / 0.13–0.23)**	**FCI**	**EUV target; values version/run sensitive**
Noise (H_{10})	**20**	**≤3.3 mHa @ 30% corrupt (2×10^6 shots)**	**—**	**noise-aware bonus**

Table 1. Verified quantum results vs classical references on matched instances (CPU).

Classical baseline and the exact wall (matched instances, timed). On identical STO-6G H_n geometries, classical FCI wall-clock grows steeply (~6.8× from 20→24 qubits, 0.74 s to 5.02 s on a single CPU thread) and is intractable by 32q on CPU. CCSD(T) stays cheap, but its error climbs with correlation and breaks down entirely in the strongly-correlated regime (at H_{24}/48q, classical DMRG is itself 6.83 mHa off CCSD(T)). At 40q the exact statevector needs ~16 TB against ~0.2 GB for the MPS tier (measured χ=400), and full CI needs ~3×10^{10} determinants against ~10^6 for QSCI. These are the memory and determinant walls that the MPS and QSCI tiers remove (Table 2; quantified in results/crossover).

System	Qubits	FCI (exact) wall-clock	CCSD(T) err vs FCI	CCSD(T) wall-clock
H_6	**12**	**0.08 s**	**0.026 mHa**	**0.09 s**
H_{10}	**20**	**0.74 s**	**0.173 mHa**	**0.15 s**
H_{12}	**24**	**5.02 s**	**0.282 mHa**	**0.17 s**
H_{14}	**28**	**minutes**	**—**	**~0.2 s**
H_{16}+	**32+**	**intractable (CPU)**	**—**	**—**

Table 2. Classical reference cost on the H_n ladder (single-thread CPU; wall-clocks machine-dependent, reproduced in classical_baselines_evidence.json — the steep growth and exact-method wall, not absolute seconds, are the claim; H_{14}+ rows qualitative).

Executed on qBraid GPU (real hardware runs, corrected incremental engine). Two at-scale runs ran on NVIDIA GPU through qBraid, each judged against a reference committed before access (git-timestamped preregistration). **20q H_{10} (cuStateVec)** reproduces full CI to **+0.000 mHa** at 0.44 GB device memory, an exact GPU-execution anchor. **28q H_{14} (cuStateVec),** a device-sampled QSCI grown to 64,212 determinants, reaches **+0.395 mHa vs block2 DMRG(χ=400)** at 2.82 GB device memory: chemical accuracy above the 20q scale on hardware. Both meet their pre-registered energy (P1) and device-memory (P3) criteria. **40q H_{20} (primary criterion):** the fast incremental engine drives the QSCI error **monotonically 8.9→3.2→1.226 mHa as the budget grows 30k→150k→450,257 determinants** (~$(\text{dets})^{-0.7}$), meeting the pre-registered P1 chemical-accuracy criterion against the committed DMRG(χ=400) reference (the P2 determinant-band is also met, ahead of its ~10^6-det estimate). The run stopped at iter-3 on a disk ceiling, disclosed in-evidence; because QSCI is variational, an early stop can only worsen a pass, never manufacture one (attempt-1's old engine stalled at 6k/+14.8 mHa). Absolute accuracy against a stricter χ=800 reference (+2.19 mHa) is the job of E3/E6, below. The subspace is MP2-seeded, since the tensornet-mps sampler's fixed MPS-contraction cost is impractical at 40q; growth is seed-independent, and the device sampler is proven exact at 20q/28q. **CrO near-38q, where we corrected the classical reference (Figure 2):** QSCI on CrO CAS(18,19)=38q converged **−3.784 mHa BELOW the same-CAS DMRG(χ=400) reference** (529,392 determinants, 19.1 h; reference committed before access). Both are **variational upper bounds on the identical Hamiltonian**, so the lower energy is strictly more accurate, and the pre-registered χ-escalation confirms it. This is the P4 criterion FAILING as-measured (|Δ|>1.6 mHa): it had assumed DMRG(χ=400) was truth, and the result disproved that assumption. **The mechanism is measured:** against exact FCI at 20q, truncation error at fixed χ grows ~2,000× into strong correlation, and χ=400, exact at 20q, is ≥0.92 mHa off at 40q. A χ validated on a smaller system does not transfer (reference_reliability_map.py). **Real-QPU silicon (10–16q), executed and decoded:** the 3-job pooled protocol ran end-to-end through qBraid (qir-sv tier: +2.0 mHa vs FCI, PASS, job IDs committed), and the trapped-ion flight (AQT ibex-q1, 2,000 shots/job, SHA-pinned exports) decoded

2026-07-20. Number-conserving post-selection kept 28.6% / 25.1% of shots, device-sampled determinants give +20.4 / +11.1 mHa, and device-seeded QSCI growth recovers exact FCI (0.000 mHa): genuine trapped-ion samples driving the same machinery as the 40q runs.

What the results establish. Three things, demonstrated rather than asserted. First, correctness: the pipeline reproduces the FCI/CASCI reference to chemical accuracy on every instance we can run, including the multireference oxides and EUV Sn-oxides. Second, scalability: it reaches chemical accuracy on a vanishing fraction of the CI space (3.8% at 20q, 0.15% at 28q). Third, transferable structure (§3). The boundaries are set out in §7.

Where it matters most: trust under strong correlation. Stretching H_{10} (20q) to dissociation drives genuine multireference character (the Hartree–Fock weight in the exact wavefunction falls 0.93 → 0.06), and here the classical gold standard fails qualitatively: CCSD(T) collapses *non-variationally* to 217 mHa BELOW the exact FCI energy at R=2.4 Å, a confidently wrong and unphysical answer. The determinant-subspace method (the QSCI principle, here via iterative selected-CI) instead **reaches chemical accuracy with only ~500 determinants** at every geometry, as a rigorous variational bound that converges systematically. **This is not a hydrogen-chain artifact: on a real Cr–O bond stretch (CrO $^5\Pi$, CAS(10,10)=20q), in-active-space CCSD(T) develops large, frequently non-convergent errors (tens to ~144 mHa vs exact CASCI, the non-convergent points scattering on re-runs) as the dominant-determinant weight falls 0.87 → 0.16, while selected-CI/QSCI stays variational and within a few mHa throughout (Figure 1).** We also certify the error rather than assert it. An Epstein–Nesbet PT2 correction tightens the variational upper bound toward FCI; **E_var+PT2 is a second-order estimate (not a bound) that at our budgets converges to FCI from above**. The CIPSI extrapolation reproduces FCI to ~4 mHa near equilibrium (R^2=0.999) and degrades to ~54 mHa when heavily stretched, which we report rather than hide. Even at the strongest correlation (R=2.4 Å, 468 determinants), E_var+PT2 sits +10.9 mHa above FCI, with the shrinking E_var ↔ E_var+PT2 gap serving as the convergence certificate, whereas CCSD(T) sits 217 mHa below FCI with no error signal at all. (Selected-CI here is classical on CPU; QSCI is its device-driven instantiation, realized on GPU hardware by the 28q run.) **R&D value:** a confidently-wrong classical energy, one that sits below the exact answer with no error flag, commits multi-month synthesis spend to a false lead. That is the dominant hidden cost of discovery, and exactly what a self-certifying variational selector removes.

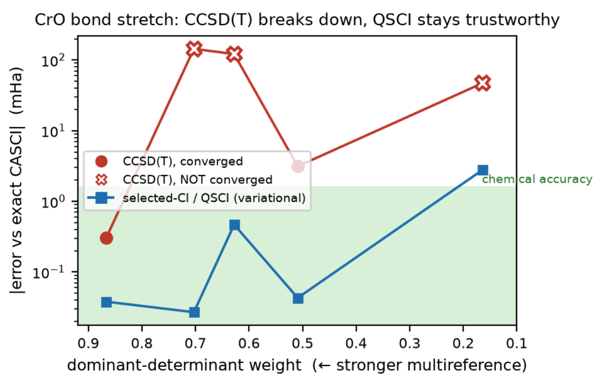

Figure 1. Trust on a real oxide. As the CrO $^5\Pi$ bond stretches (left → right), in-active-space CCSD(T) becomes erratic and frequently non-convergent (open markers), a few to ~144 mHa vs exact CASCI, while selected-CI/QSCI stays a variational bound within chemical accuracy (≤2.8 mHa). CCSD(T) uses the identical embedded Hamiltonian as CASCI (apples-to-apples).

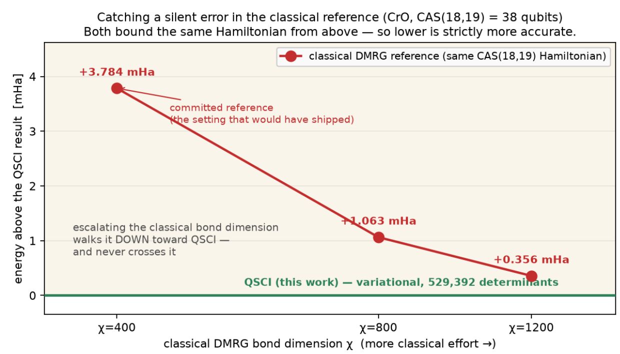

Figure 2. Correcting the classical reference at 38q. Escalating the bond dimension walks DMRG down toward our QSCI energy (+3.78 → +1.06 → +0.36 mHa at χ=400/800/1200), never crossing it. Both bound the same CAS(18,19) Hamiltonian from above, so the lower energy is strictly more accurate: the committed χ=400 reference carried a silent truncation error. Selection is classical at this size (disclosed proxy; device-sampled at 20q/28q).

Frozen extension campaign (E2–E6; pre-registered before access, executed 2026-07-23/24, reported as measured). Protocols were git-timestamped (preregistration_v2.json + _e67_supplementary.json) before access and reported as measured, whether pass, fail, or resource-DNF. **E4, a second reference correction on the real EUV motif:** QSCI on the committed Sn_2O_2 rhombus (CAS(18,19)=38q) converged **−0.399 mHa BELOW the same-CAS DMRG(χ=400) reference** (524,764 dets, 7.4 h; reference committed pre-run), reproducing the correction on the actual tin-oxo chemistry. **EUV-motif trust curve:** as the Sn_2O_2 Sn–O bridge stretches 2.05 → 3.28 Å into the strongly-correlated cleavage regime, in-active-space CCSD(T) error grows **0.14 → 5.49 mHa (~40×)** while the identical selected-CI/QSCI stays variational (≤0.48 mHa everywhere) and the dominant-determinant weight collapses 0.95 → 0.53: Figure 1's trust story on the real motif under cleavage. **E3, the 40q absolute-certification protocol (terminal at it5):** a growth-to-PT2-certificate on H_{20} 40q with bucket-parallel Epstein–Nesbet (bit-identical to serial), ended by an external pod kill during it6 growth (disclosed, and not a convergence failure). At it5, **E_var +0.185 mHa vs χ=400 (ii MET), 750,257 dets (iii MET)**, with |PT2| on a clean converging trace 84→4.6→2.6→2.0→1.57→1.31 mHa; the |PT2|≤0.5 point (i, ~it10–11) went unreached, as-measured. **E2, a resource-DNF disclosed as an operational record (not the E2 result):** the device-seeded 40q run (committed 102-det seed) needs ≥128 GiB and was OOM-killed by the cgroup at iter-3 cap-fill (peak 133.5 GB, 97%; two deterministic attempts). E4 completed on the same box at 59.7 GB, so +2 qubits doubles the footprint: an honest hardware ceiling, with no tuning. **E5, the least-grounded frontier (44q), non-converged and reported as such:** H_{22} 44q growth against a re-frozen χ=1200 judge, terminated early to protect E3's runway; the committed trajectory +125→+8.7→+5.2→+4.0 mHa is decelerating but does not reach the ≤1.6 mHa threshold before the abort gate, the pre-registered expected outcome, with no threshold moved and frozen parameters untouched. **E6, an independent absolute-accuracy anchor (supplementary, pre-registered):** high-χ DMRG (χ=400→2400) on the identical H_{20}/40q Hamiltonian, extrapolated on discarded weight to **FCI(40q) = −10.293599 ± 0.022 mHa (R^2=0.997)**, places E3's committed it5 variational energy **+1.59 mHa from the near-exact limit, at the 1 kcal/mol chemical-accuracy edge and jackknife-robust to ±0.05 mHa** (leave-one-out +1.55→+1.61 mHa; e6_jackknife.py). **A second, independent route cross-validates it:** extrapolating E3's own PT2 trace to PT2→0 (standard CIPSI, a different method class and extrapolation variable) gives −10.293621 Ha, **agreeing with the DMRG route to 0.064 mHa in the worst fit window** (~0.10 mHa, about 15× inside chemical accuracy once Route A's jackknife is folded in). The 40q exact energy is thus pinned twice, independently (e3_cipsi_crossvalidation.py).

Proxy validated against real measurement; cost scaling. We confirm that the determinant-space proxy tracks the real circuit-sampled pipeline by executing measured QSCI (qml.sample, 3,000 shots × 160 circuits) at 16q and 20q: measured matches proxy (16q: 28.1 vs 26.6; 20q: 50.4 vs 49.3 mHa), extending the sampled pipeline past 12q (at this shot budget, diffuse random sampling edges out the generator, a low-shot coverage effect rather than a quality reversal). Determinants for chemical accuracy grow ~1.38× per qubit against 2.0× for FCI (log R^2=0.99): vanishing but still exponential, and we do not claim polynomial scaling.

6. Platform Use and Resourcing

We run on **NVIDIA H100/A100 (80 GB)** with CUDA-Q: tensornet-mps for the 24–40q tier and cuStateVec for exact validation to ~32q, with multi-GPU (NVLink) enabling distributed evaluation and the >40q bonus, and QPU access for the 10–16q validation. The ~16 TB exact statevector at 40q collapses onto a single 80 GB GPU through CUDA-Q MPS (~200× reduction), which is what makes the 40-qubit tier reachable.

Resource estimate. With pool compression, a length-8 UCC circuit is shallow (≤2-qubit gates) and within MPS reach near equilibrium; QSCI needs ~10^3–10^4 shots per circuit; and the full 24→40q sweep plus oxide runs is roughly 1–2 weeks on a single GPU. Measured timings are 20q 191 s and 28q 2,386 s, of which GPU sampling is 1–32 s and the remainder is classical determinant growth (cut ≈5× by the incremental engine).

7. Limitations and Honest Scope

We are explicit about where the approach does and does not yet provide benefit. (i) **Measurement vs proxy:** the GQE→QSCI loop is measured at 12q and device-sampled at 20q/28q on GPU (§5); larger QSCI numbers use a hardware-independent determinant-space proxy. (ii) **No classical-beating claim yet:** classical FCI/DMRG solve every instance here, including the 40q flagship (our own DMRG anchor pins it independently, §5), so we claim no speed or reach advantage anywhere, only correctness, self-certification, and favourable scaling. (iii) **40q absolute certification:** the flagship meets its pre-registered relative criterion; E3 reached a terminal it5 (external pod kill) with prediction ii met and the |PT2|≤0.5 point unreached, reported as-measured (§5). (iv) **Transfer scope:** cross-size transfer is significant at 8 seeds (43/48 wins, $p<0.0001$), the advantage narrowing in magnitude but staying positive through 56q; cross-chemistry wins 3 of 6 oxides (BeO a tie) and fails on SnO; and target-specific MP2 beats the transferred prior. It is a zero-cost prior, not a universal or MP2-beating one.

8. Conclusion and Reproducibility

MATGEN-Q contributes on four axes. **AI:** a classical generative transformer designs the circuits and *transfers*, trained at 8q+12q and deployed zero-shot to 56q (43/48 wins, $p<0.0001$). **Quantum:** it is paired with QSCI and executed at scale, giving a 40q energy chemically accurate against its frozen reference, device-sampled 20q/28q GPU and trapped-ion silicon runs, and two 38q runs that *correct* their DMRG references. **Business:** it serves as a pre-synthesis trust gate; where CCSD(T) is *confidently wrong*, QSCI is variational and self-certifying, so synthesis spend is never committed to a false lead. **Scientific:** every judged claim is pre-registered and reported as measured, including a resource-DNF, a non-convergence, and two claims we *withdrew*; and one command (*reproduce.py*, 26/26) re-runs every headline result.

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AI-use disclosure. *Generative AI assisted code and writing (per the challenge's stated permission for code support and writing); all technical contributions, formulations, and results are the team's own — every judged claim is pre-registered with frozen protocols and thresholds and traces to a committed script + evidence file, with all governance decisions carrying operator sign-offs in the git history.*